The **problem statement** is to build a machine learning model that can predict heart disease based on patient health data.

Objective

1. To analyze patient health data
2. To build and compare machine learning models for heart disease prediction
3. To evaluate model performance using appropriate metrics
4. To select the best-performing model for prediction

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

import warnings
warnings.filterwarnings("ignore")
```

In [2]:
```
df = pd.read_csv("heart.csv")
```

In [3]:
```
df.head()
```

Out[3]:

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target
0	63	1	3	145	233	1	0	150	0	2.3	0	0	1	1
1	37	1	2	130	250	0	1	187	0	3.5	0	0	2	1
2	41	0	1	130	204	0	0	172	0	1.4	2	0	2	1
3	56	1	1	120	236	0	1	178	0	0.8	2	0	2	1
4	57	0	0	120	354	0	1	163	1	0.6	2	0	2	1

In [4]:
```
# Rows - 303
# column - 14
df.shape
```

Out[4]: (303, 14)

In [5]:
```
# non null values are present in columns so need of null processing
# object column is not present
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 303 entries, 0 to 302
Data columns (total 14 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   age       303 non-null    int64
 1   sex       303 non-null    int64
 2   cp        303 non-null    int64
 3   trestbps  303 non-null    int64
 4   chol      303 non-null    int64
 5   fbs       303 non-null    int64
 6   restecg   303 non-null    int64
 7   thalach   303 non-null    int64
 8   exang     303 non-null    int64
 9   oldpeak   303 non-null    float64
 10  slope     303 non-null    int64
 11  ca        303 non-null    int64
 12  thal      303 non-null    int64
 13  target    303 non-null    int64
dtypes: float64(1), int64(13)
memory usage: 33.3 KB
```

In [6]:
```
# checking null values
# no missing values are present
df.isnull().sum()
```

Out[6]:

```
age         0
sex         0
cp          0
trestbps    0
chol        0
fbs         0
restecg     0
thalach     0
exang       0
oldpeak     0
slope       0
ca          0
thal        0
target      0
dtype: int64
```

In [7]:

```
# Right skewed - trestbps, chol,
# left skewed - thalach
# little bit skewed data is present
df.describe()
```

Out[7]:

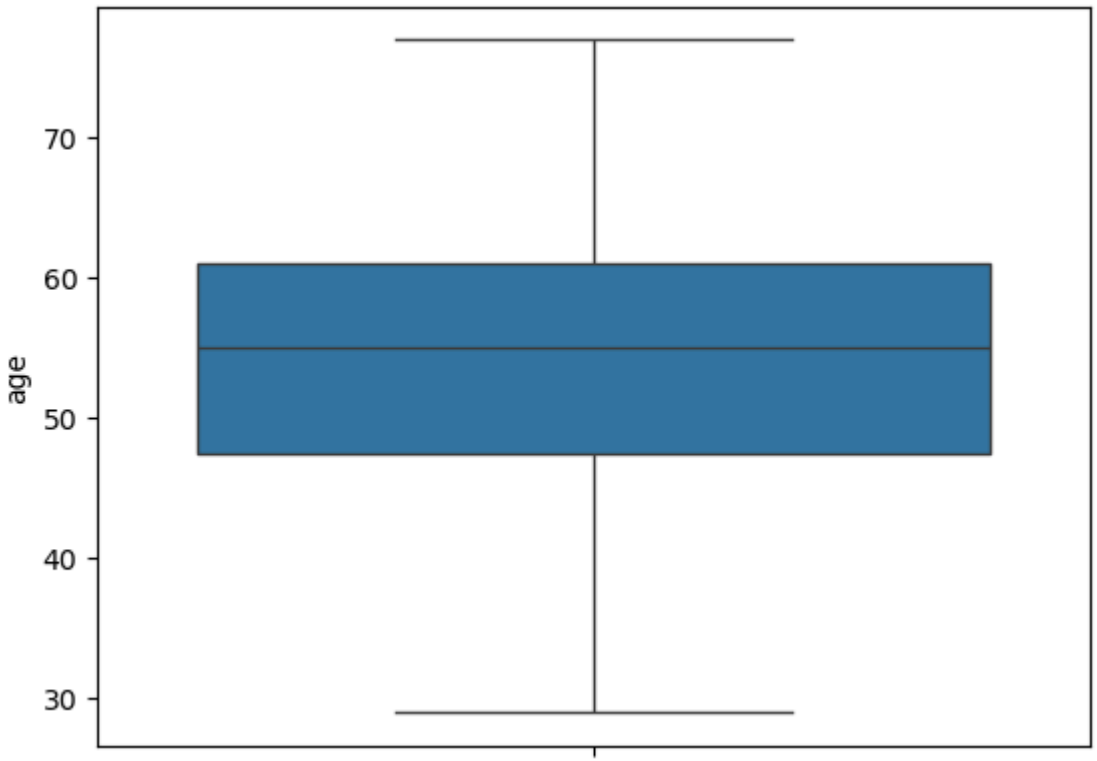
	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope
count	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000
mean	54.366337	0.683168	0.966997	131.623762	246.264026	0.148515	0.528053	149.646865	0.326733	1.039604	1.399340
std	9.082101	0.466011	1.032052	17.538143	51.830751	0.356198	0.525860	22.905161	0.469794	1.161075	0.616226
min	29.000000	0.000000	0.000000	94.000000	126.000000	0.000000	0.000000	71.000000	0.000000	0.000000	0.000000
25%	47.500000	0.000000	0.000000	120.000000	211.000000	0.000000	0.000000	133.500000	0.000000	0.000000	1.000000
50%	55.000000	1.000000	1.000000	130.000000	240.000000	0.000000	1.000000	153.000000	0.000000	0.800000	1.000000
75%	61.000000	1.000000	2.000000	140.000000	274.500000	0.000000	1.000000	166.000000	1.000000	1.600000	2.000000
max	77.000000	1.000000	3.000000	200.000000	564.000000	1.000000	2.000000	202.000000	1.000000	6.200000	2.000000

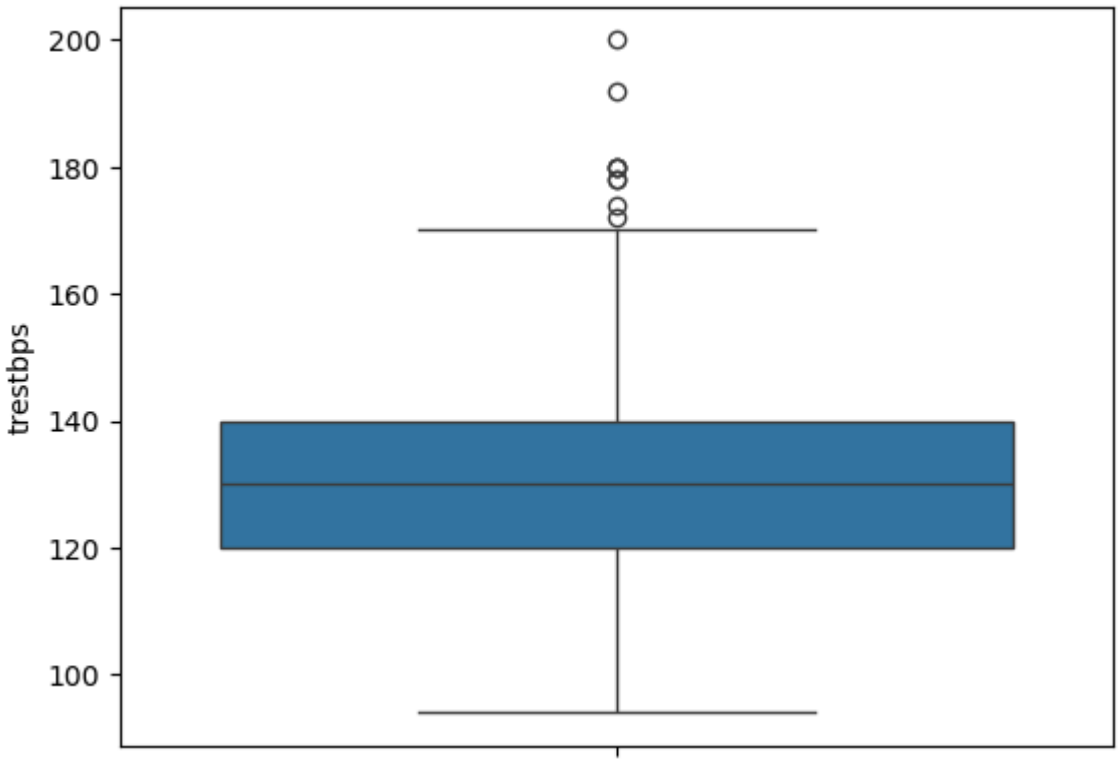
In [8]:
```
# data is not ither banced or embanced
df["target"].value_counts()
```

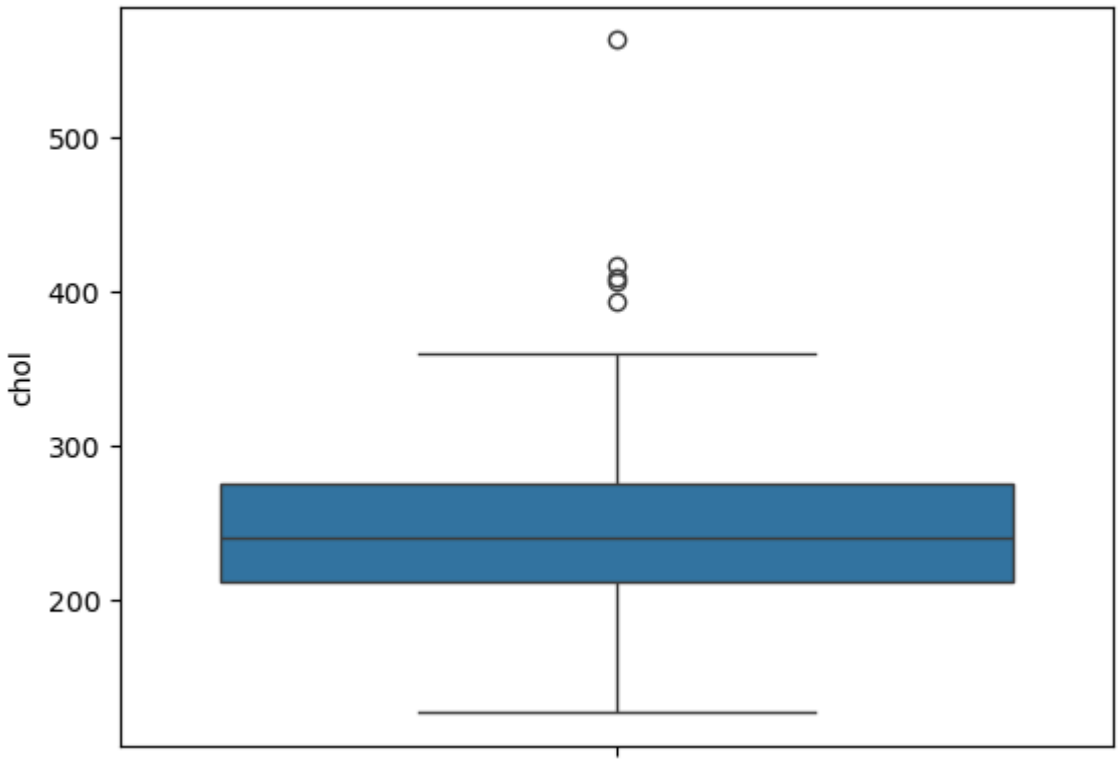
Out[8]:
```
target
1    165
0    138
Name: count, dtype: int64
```

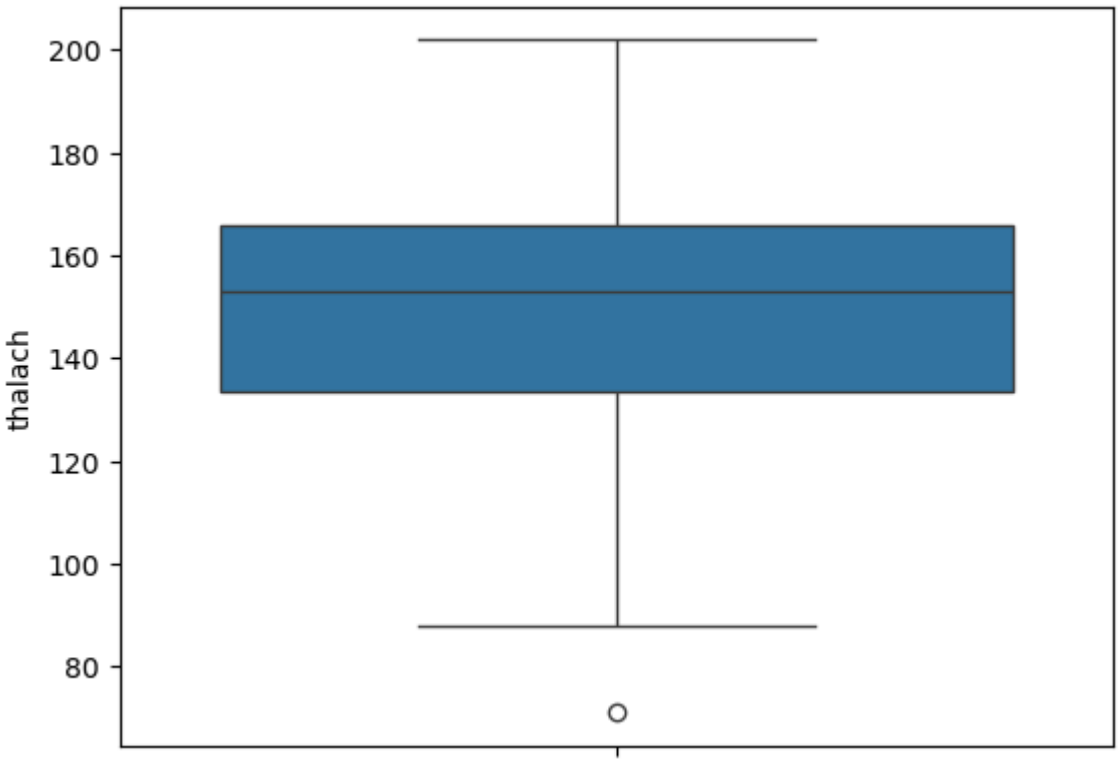
In [9]:
```
# seperating the numerical columns for finding the outliers
num_col = ['age','trestbps','chol','thalach','oldpeak']
```

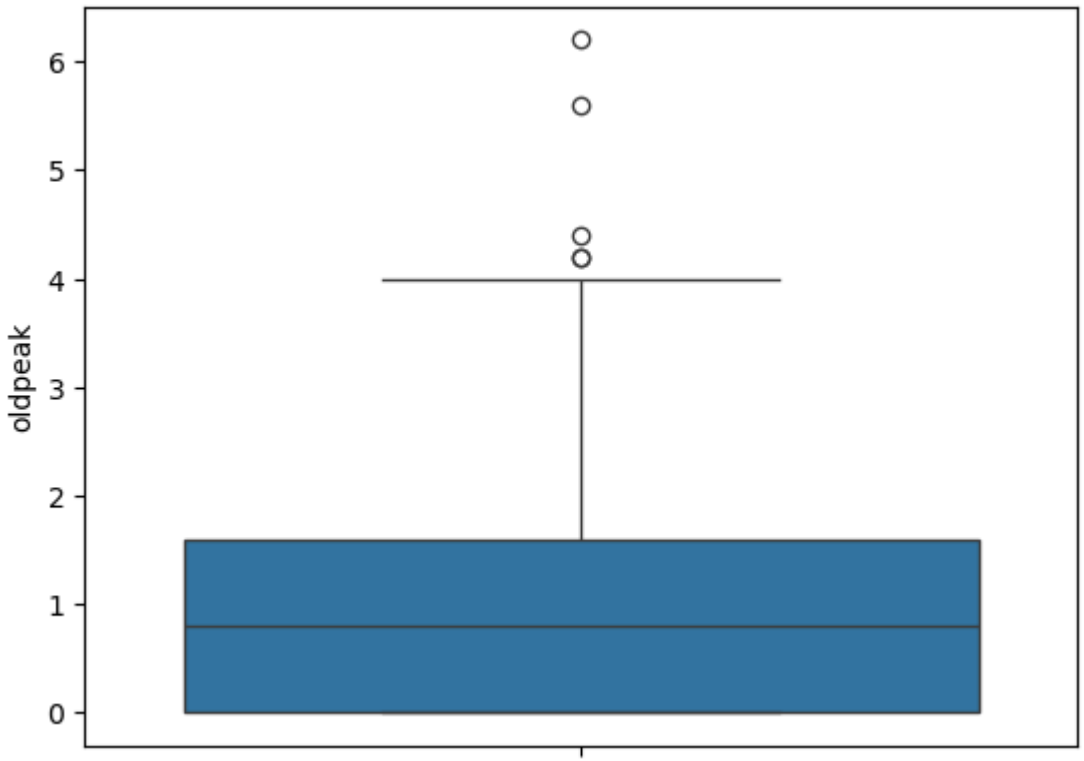
In [10]:
```
# some columns like trestbps,chol,thalach,oldpeak contains the outlier but i am not removing them according to domain knowledg

for col in num_col:
    plt.figure()
    sns.boxplot(df[col])
    plt.show()
```


70
60
50
40
30
age

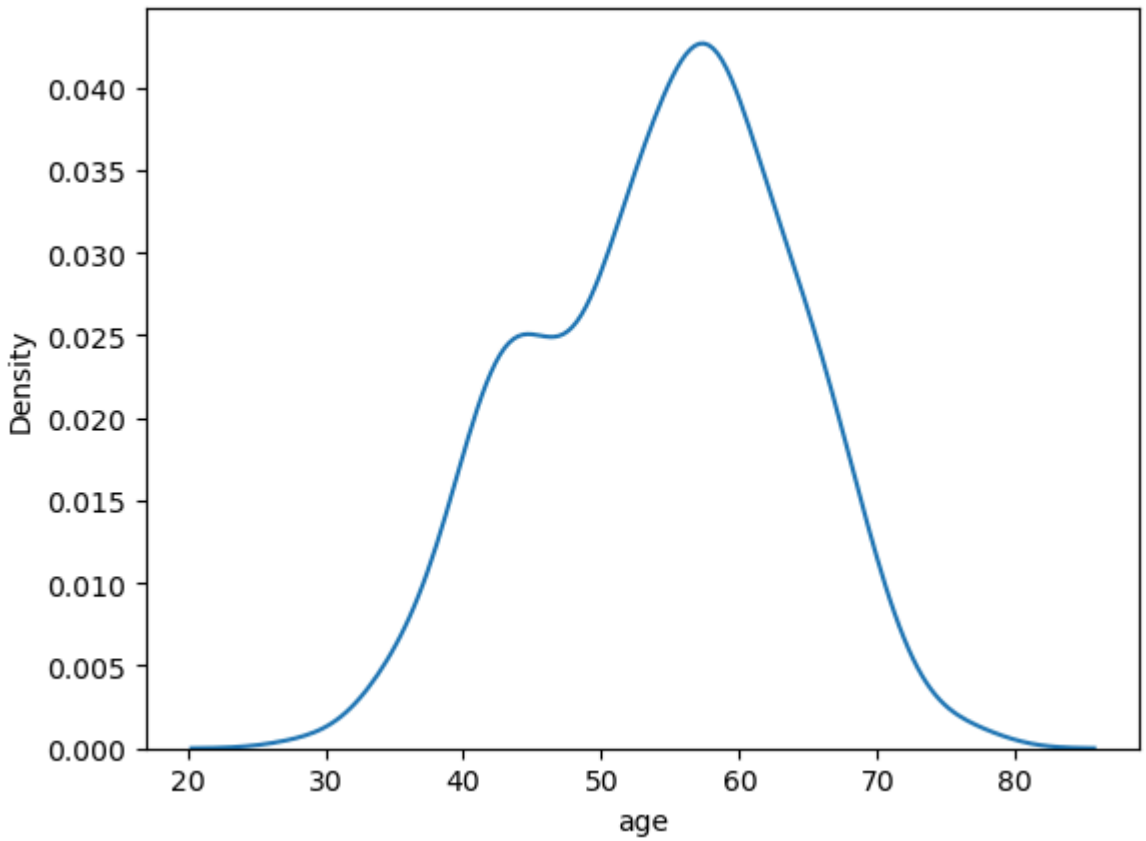

200
180
160
140
120
100
trestbps

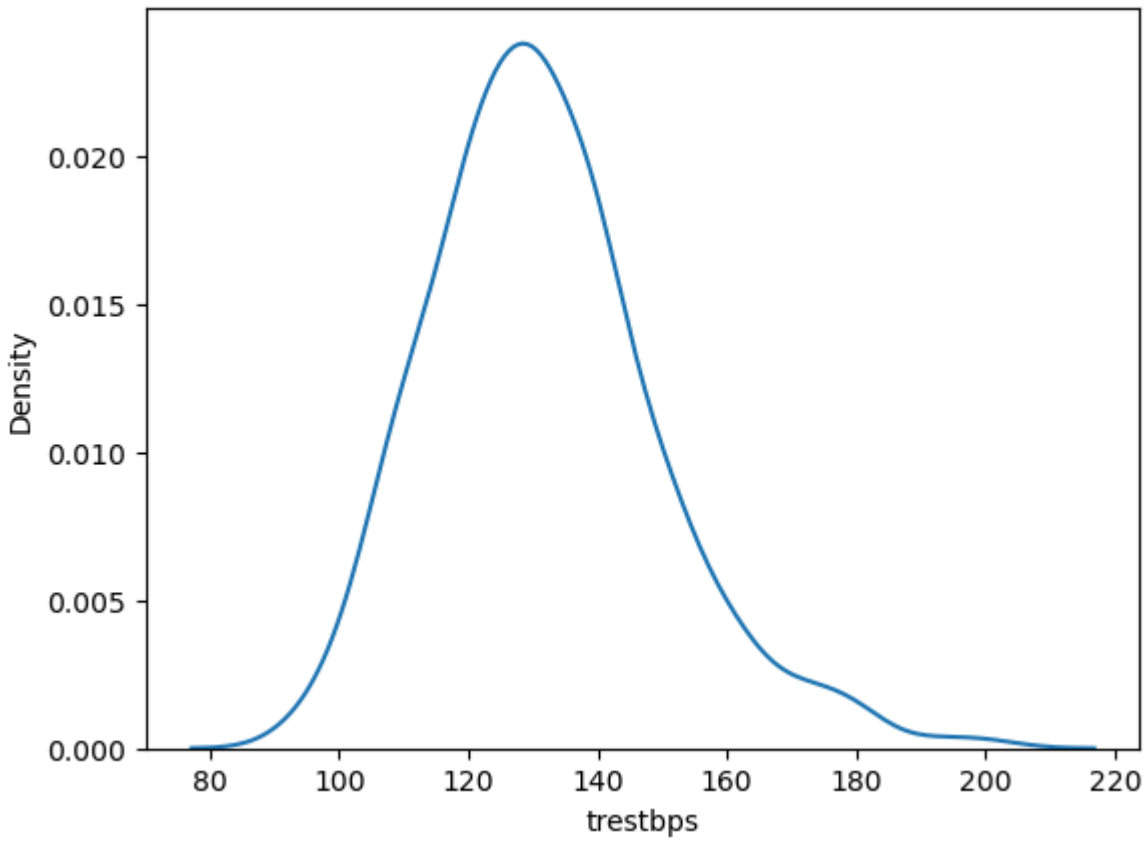

500
400
300
200
chol

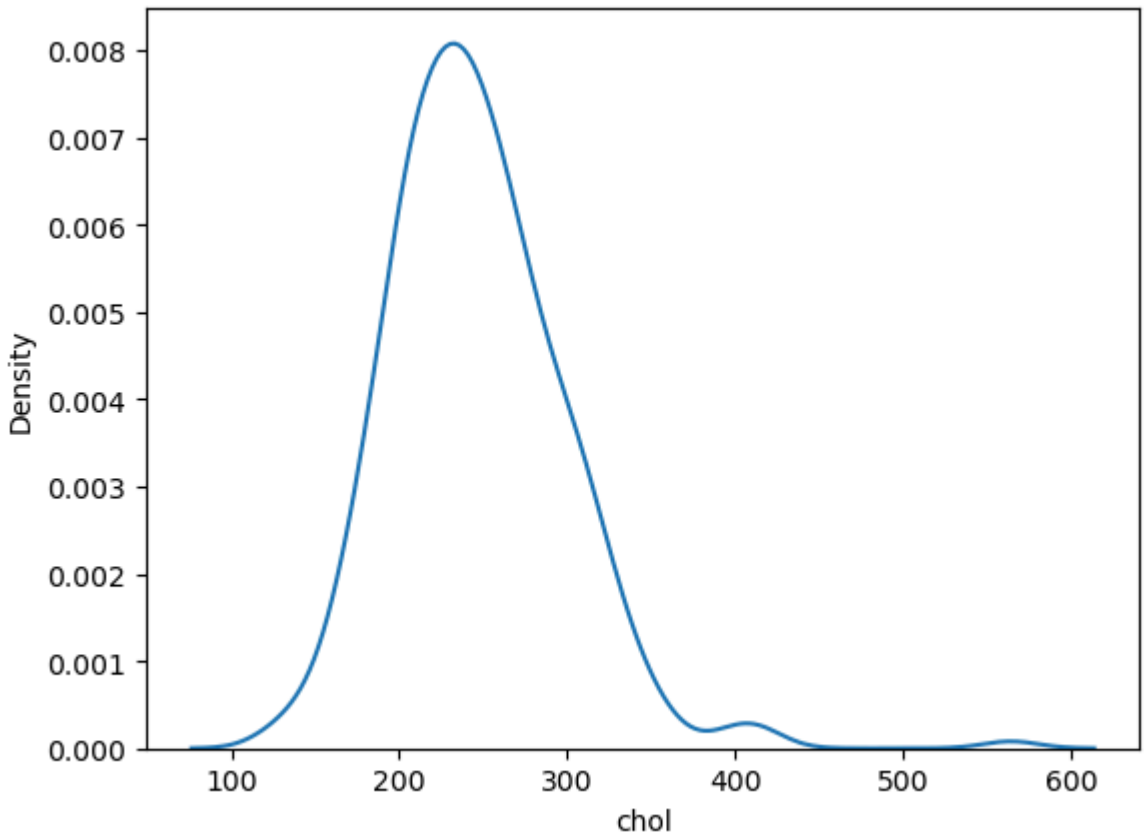

200
180
160
140
120
100
80
thalach

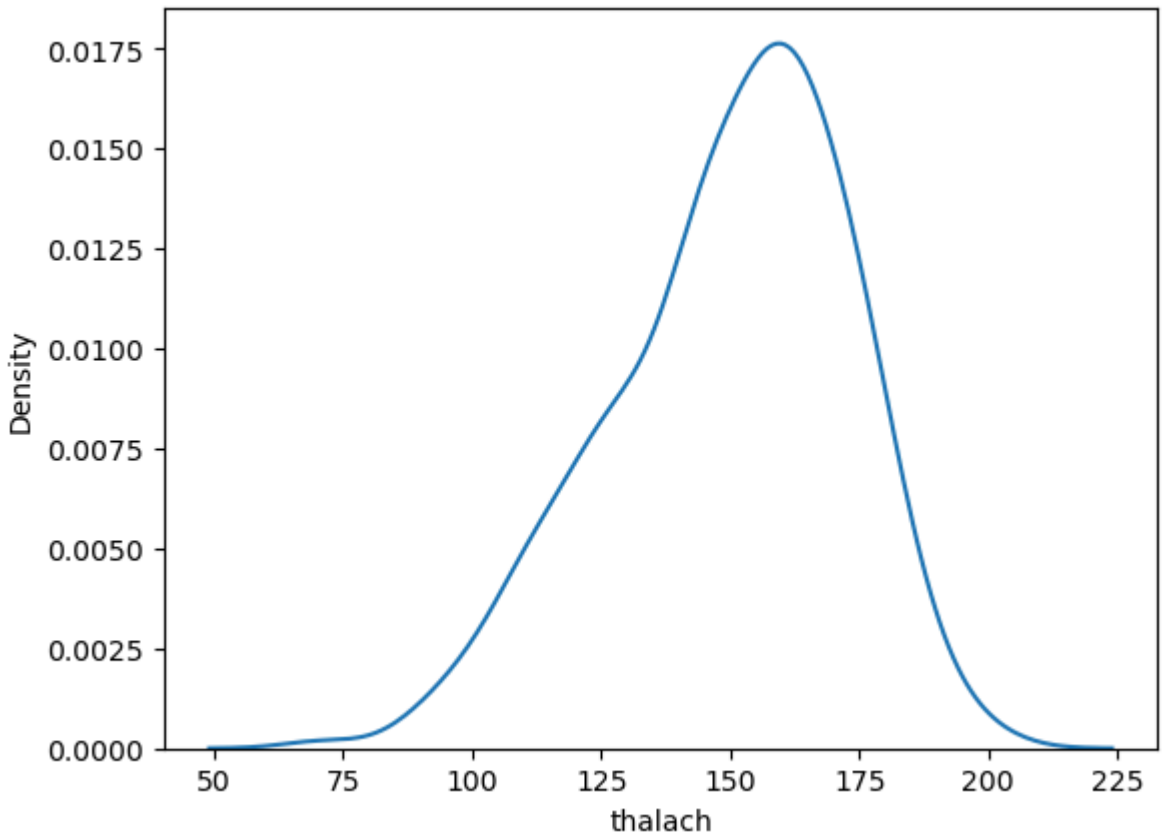


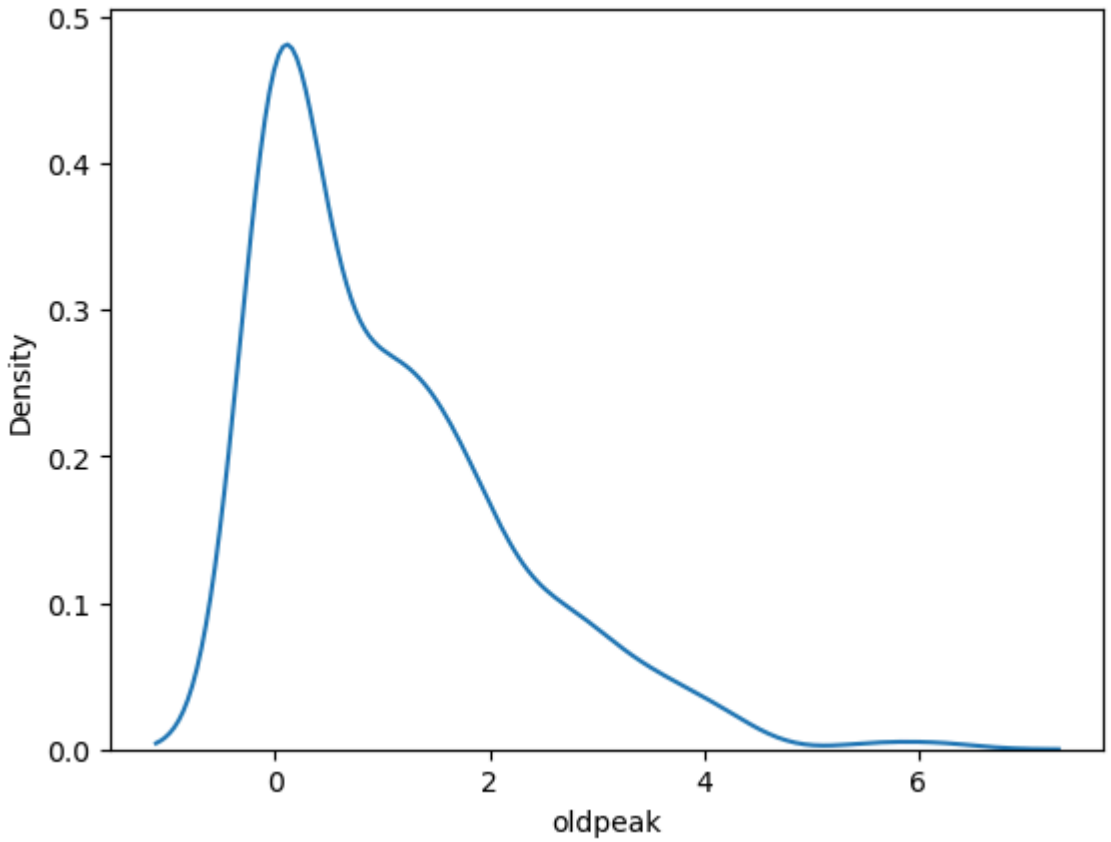
In [11]:
```
# all columns are highly skewed
# left skewness - age, thalach
# right skewness - trestbps, chol, oldpeak
# according to my understanding i am not removing the skewness because it directly impact on my features
for col in num_col:
    plt.figure()
    sns.kdeplot(df[col])
    plt.show()
```


0.040
0.035
0.030
0.025
0.020
0.015
0.010
0.005
0.000
Density
20
30
40
50
60
70
80
age

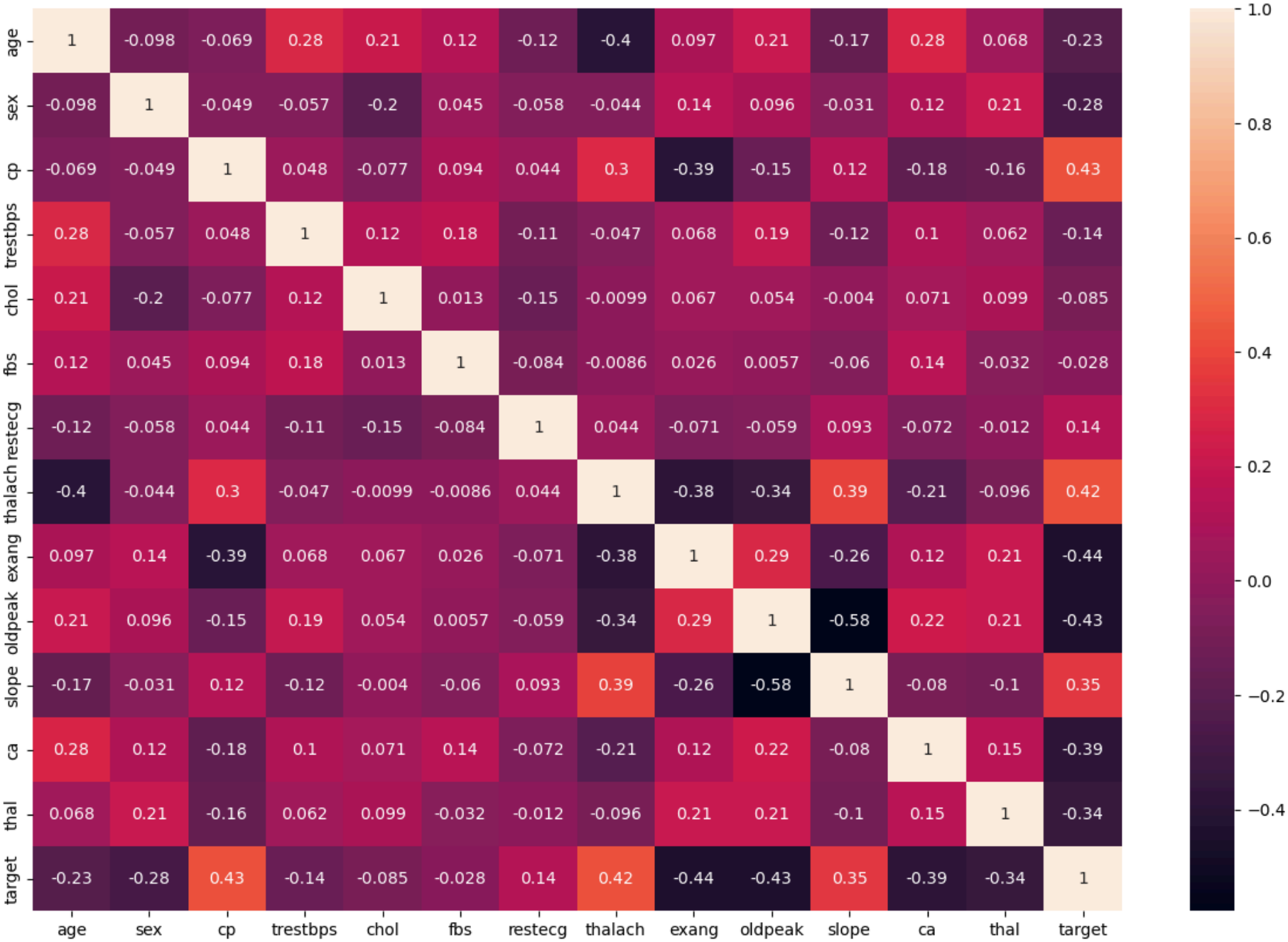

0.020
0.015
0.010
0.005
0.000
Density
80
100
120
140
160
180
200
220
trestbps


0.008
0.007
0.006
0.005
0.004
0.003
0.002
0.001
0.000
Density
100
200
300
400
500
600
chol

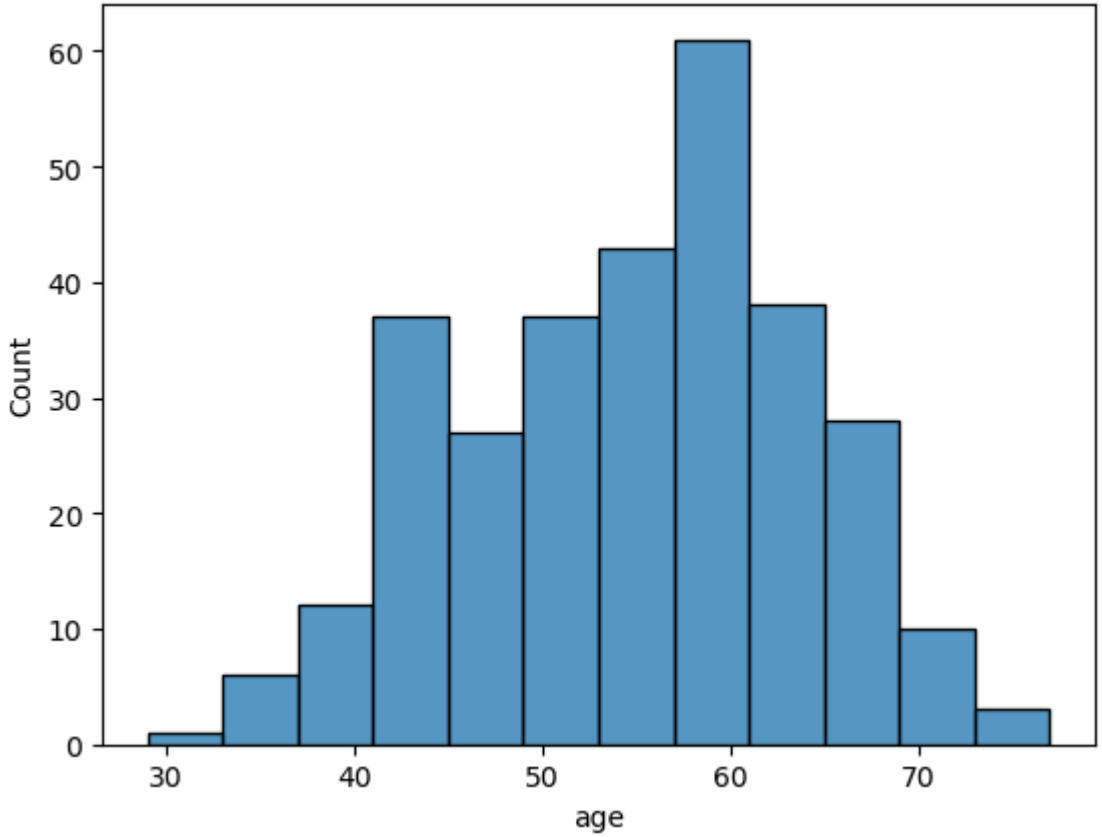

0.0175
0.0150
0.0125
0.0100
0.0075
0.0050
0.0025
0.0000
Density
50
75
100
125
150
175
200
225
thalach

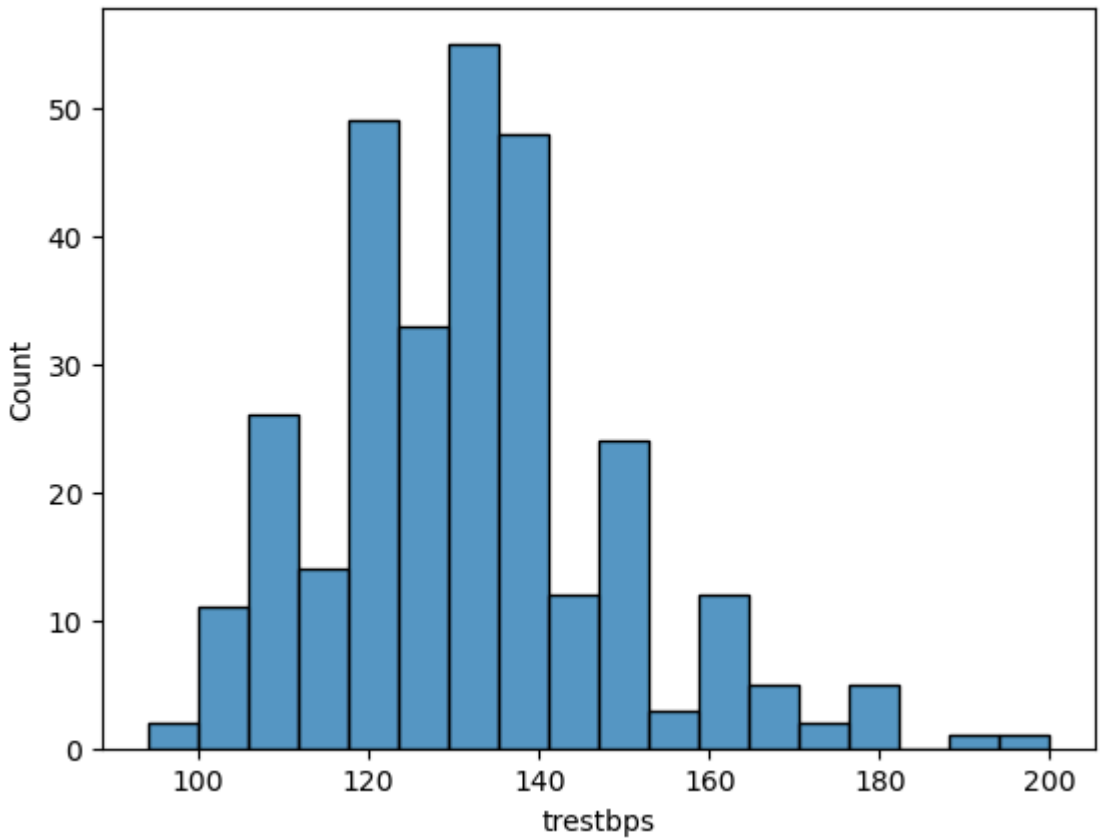


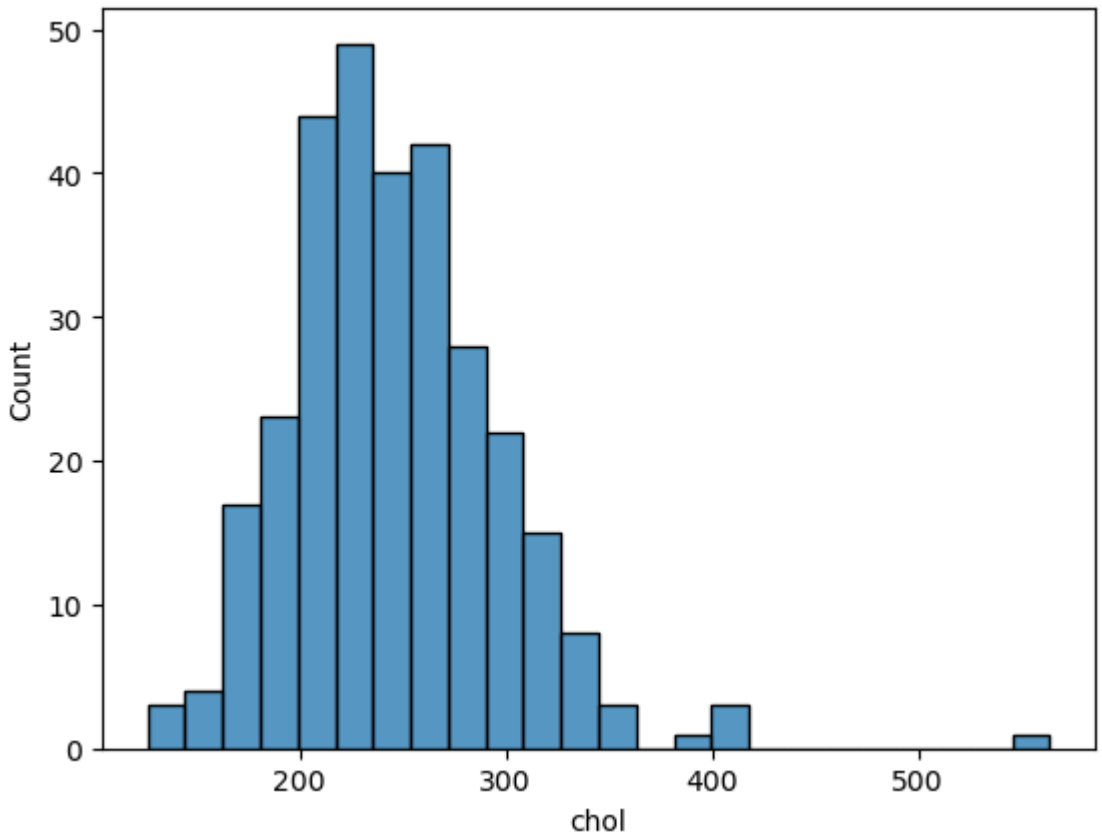
In [12]:
```
# cp and slope are highly correlated with target column
plt.figure(figsize = (15,10))
sns.heatmap(df.corr(), annot = True)
plt.show()
```

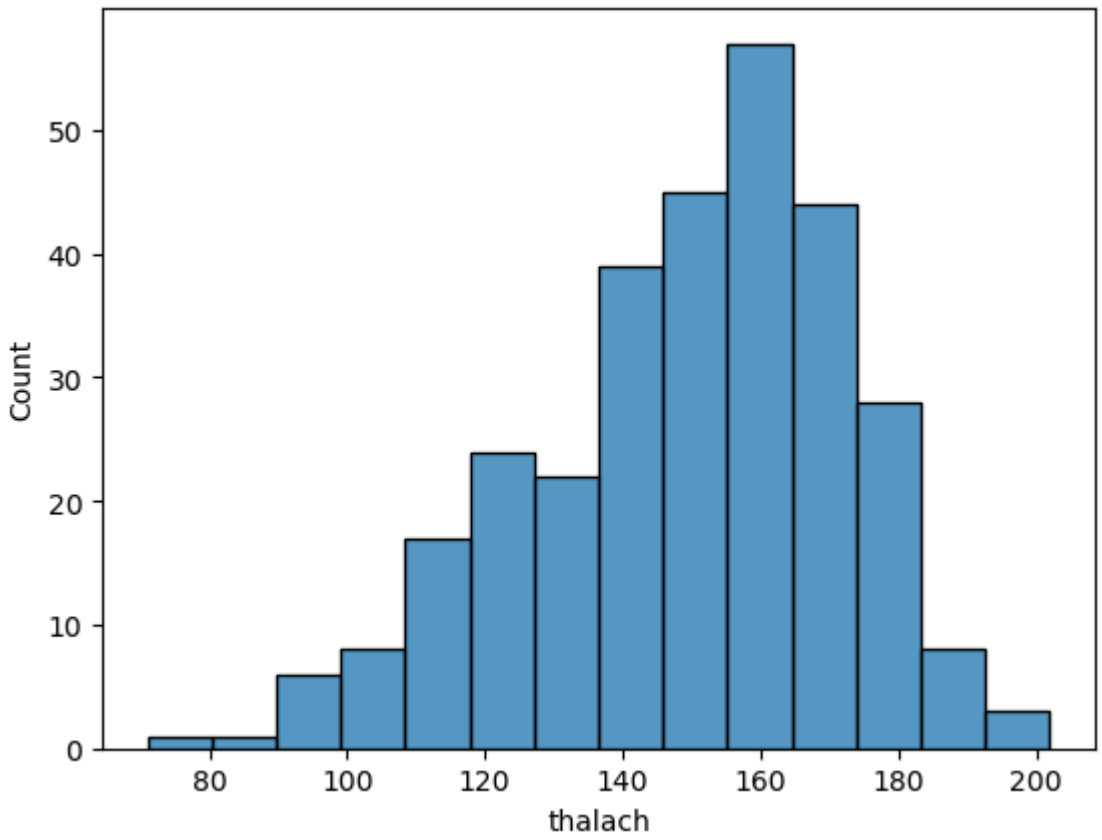

age 1 -0.098 -0.069 0.28 0.21 0.12 -0.12 -0.4 0.097 0.21 -0.17 0.28 0.068 -0.23
sex -0.098 1 -0.049 -0.057 -0.2 0.045 -0.058 -0.044 0.14 0.096 -0.031 0.12 0.21 -0.28
cp -0.069 -0.049 1 0.048 -0.077 0.094 0.044 0.3 -0.39 -0.15 0.12 -0.18 -0.16 0.43
trestbps 0.28 -0.057 0.048 1 0.12 0.18 -0.11 -0.047 0.068 0.19 -0.12 0.1 0.062 -0.14
chol 0.21 -0.2 -0.077 0.12 1 0.013 -0.15 -0.0099 0.067 0.054 -0.004 0.071 0.099 -0.085
fbs 0.12 0.045 0.094 0.18 0.013 1 -0.084 -0.0086 0.026 0.0057 -0.06 0.14 -0.032 -0.028
restecg -0.12 -0.058 0.044 -0.11 -0.15 -0.084 1 0.044 -0.071 -0.059 0.093 -0.072 -0.012 0.14
thalach -0.4 -0.044 0.3 -0.047 -0.0099 -0.0086 0.044 1 -0.38 -0.34 0.39 -0.21 -0.096 0.42
exang 0.097 0.14 -0.39 0.068 0.067 0.026 -0.071 -0.38 1 0.29 -0.26 0.12 0.21 -0.44
oldpeak 0.21 0.096 -0.15 0.19 0.054 0.0057 -0.059 -0.34 0.29 1 -0.58 0.22 0.21 -0.43
slope -0.17 -0.031 0.12 -0.12 -0.004 -0.06 0.093 0.39 -0.26 -0.58 1 -0.08 -0.1 0.35
ca 0.28 0.12 -0.18 0.1 0.071 0.14 -0.072 -0.21 0.12 0.22 -0.08 1 0.15 -0.39
thal 0.068 0.21 -0.16 0.062 0.099 -0.032 -0.012 -0.096 0.21 0.21 -0.1 0.15 1 -0.34
target -0.23 -0.28 0.43 -0.14 -0.085 -0.028 0.14 0.42 -0.44 -0.43 0.35 -0.39 -0.34 1
age sex cp trestbps chol fbs restecg thalach exang oldpeak slope ca thal target
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4

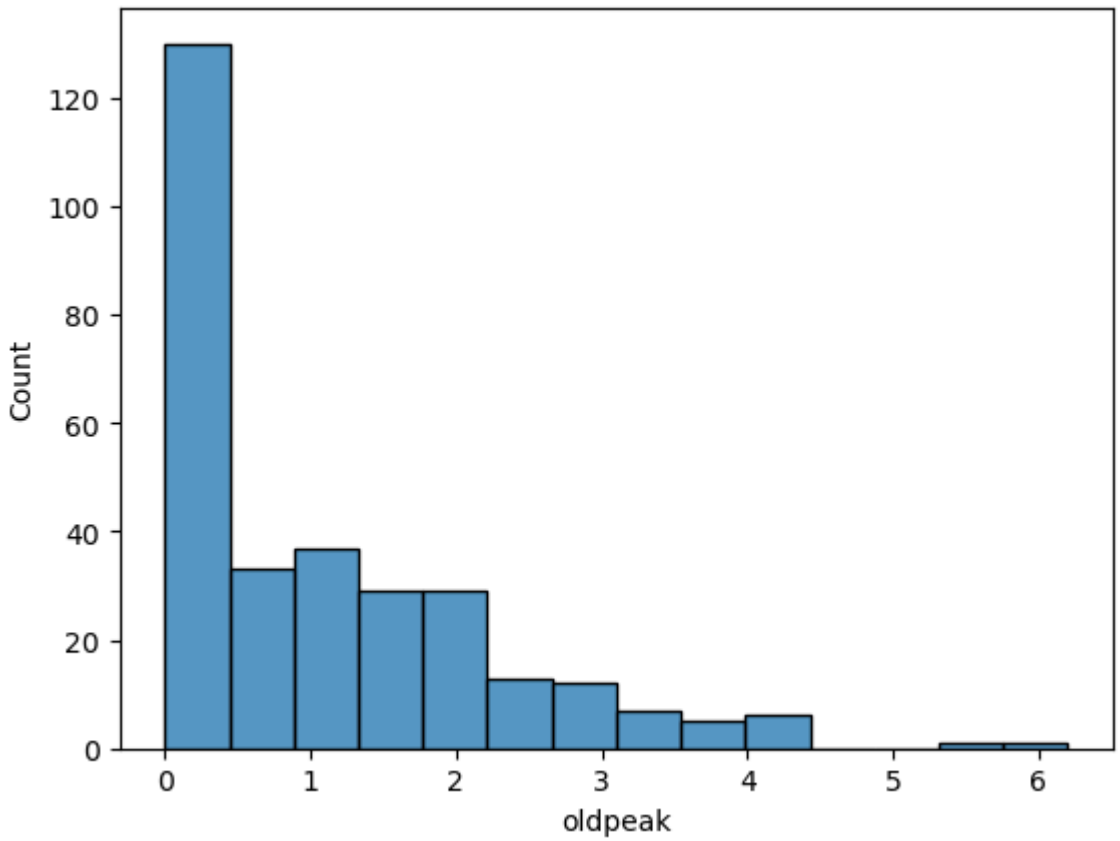
In [13]:
```
for col in num_col:
    plt.figure()
    sns.histplot(df[col])
    plt.show()
```




Count
0
10
20
30
40
50
100
120
140
160
180
200
trestbps


50
40
30
20
10
0
Count
200
300
400
500
chol


Count
0
10
20
30
40
50
80
100
120
140
160
180
200
thalach



In [14]:
```
x = df.drop("target", axis=1) # independent variables
y = df["target"] # dependent variable
```

In [15]:
```
x
```

Out[15]:

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal
0	63	1	3	145	233	1	0	150	0	2.3	0	0	1
1	37	1	2	130	250	0	1	187	0	3.5	0	0	2
2	41	0	1	130	204	0	0	172	0	1.4	2	0	2
3	56	1	1	120	236	0	1	178	0	0.8	2	0	2
4	57	0	0	120	354	0	1	163	1	0.6	2	0	2
...	...	...	...	...	...	...	...	...	...	...	...	...	...
298	57	0	0	140	241	0	1	123	1	0.2	1	0	3
299	45	1	3	110	264	0	1	132	0	1.2	1	0	3
300	68	1	0	144	193	1	1	141	0	3.4	1	2	3
301	57	1	0	130	131	0	1	115	1	1.2	1	1	3
302	57	0	1	130	236	0	0	174	0	0.0	1	1	2

303 rows × 13 columns

In [16]:
```
y
```

Out[16]:
```
0      1
1      1
2      1
3      1
4      1
      ..
298    0
299    0
300    0
301    0
302    0
Name: target, Length: 303, dtype: int64
```

In [17]:
```
x_duplicate = x
```

In [18]:
```
# scalling the independent varible for better result
from sklearn.preprocessing import StandardScaler
ss = StandardScaler()
x_scal = ss.fit_transform(x)
x = pd.DataFrame(x_scal)
```

In [19]:
```
from sklearn.model_selection import train_test_split
x_train, x_test, y_train, y_test = train_test_split(x,y,test_size = 0.30, random_state=42)
```

In [20]:
```
# x_train contains 212 rows and 13 columns for training the indipendence varibles
x_train.shape
```

Out[20]:
```
(212, 13)
```

In [21]:
```
# 91 rows and 13 columns are used for the test
x_test.shape
```

Out[21]:
```
(91, 13)
```

In [22]:
```
# y_train contains 212 rows for training dependent varible
y_train.shape
```

Out[22]:
```
(212,)
```

In [23]:
```
# 91 rows are used for the test
y_test.shape
```

Out[23]:
```
(91,)
```

Logistic Regression

In [24]:
```
from sklearn.linear_model import LogisticRegression
```

In [25]:
```
model = LogisticRegression()
```

In [26]:
```
# training data
model.fit(x_train, y_train)
```

Out[26]:
```
LogisticRegression
LogisticRegression()
```

In [27]:
```
# prediction
y_pred = model.predict(x_test)
print(y_pred)
```

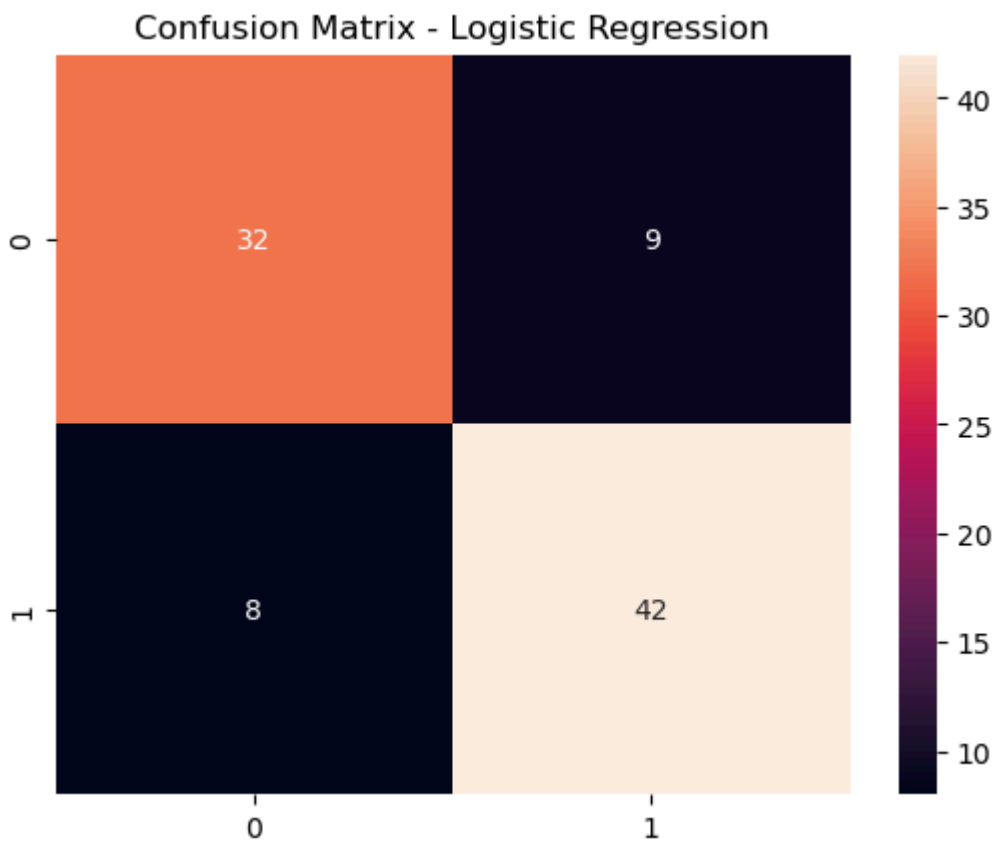
```
[0 1 1 0 1 1 1 0 0 0 1 0 1 0 1 1 1 0 0 0 1 0 0 1 1 1 1 1 0 1 0 0 0 0 1 0 1
 1 1 1 0 1 1 1 1 0 1 1 0 0 0 0 1 1 0 0 0 1 0 0 0 1 0 1 1 0 1 1 1 1 1 1 1 1
 0 1 1 1 0 0 0 0 1 1 1 0 0 1 1 0 1]
```

In [28]:
```
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix, roc_auc_score,recall_score
```

In [29]:
```
print(classification_report(y_test,y_pred))
print("roc_auc score: ",roc_auc_score(y_test,y_pred))
```

```
              precision    recall  f1-score   support

           0       0.80      0.78      0.79        41
           1       0.82      0.84      0.83        50

    accuracy                           0.81        91
   macro avg       0.81      0.81      0.81        91
weighted avg       0.81      0.81      0.81        91

roc_auc score:  0.8102439024390243
```

In [30]:
```
# false negative is dangeros for the heart desease prediction
cm = confusion_matrix(y_test,y_pred)
plt.figure()
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix - Logistic Regression")
sns.heatmap(cm, annot =True)
plt.show()
```



After implementing the **loggistic regression** model i get 81% accuracy

Decision Tree Classifier

In [31]:
```
# Decision Tree is used because it is simple, easy to understand, and makes decisions using if-else conditions.
from sklearn.tree import DecisionTreeClassifier, plot_tree
```

In [32]:
```
dt = DecisionTreeClassifier()
```

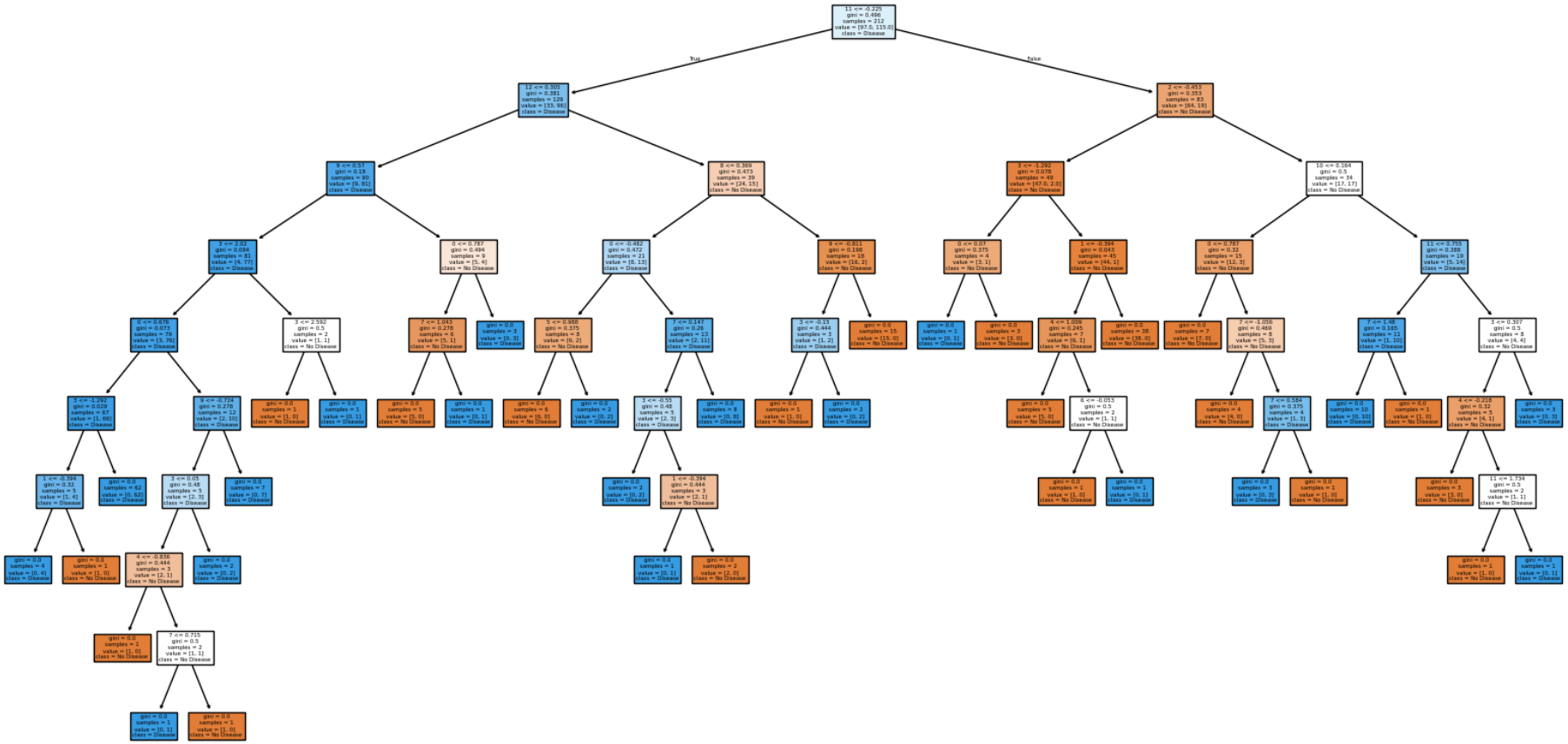
In [33]:
```
dt.fit(x_train,y_train)
```

Out[33]:

```
DecisionTreeClassifier
DecisionTreeClassifier()
```

In [34]:

```
plt.figure(figsize = (20,10))
plot_tree(dt, feature_names = x.columns, class_names = ["No Disease", "Disease"],filled = True)
plt.show()
```



In [35]:

```
dt_pred = dt.predict(x_test)
print(dt_pred)
```

```
[0 1 1 0 1 1 1 0 0 0 1 0 1 0 1 1 1 0 0 0 1 0 1 1 0 0 1 1 0 1 0 0 0 0 1 0 0
 0 0 1 1 1 1 0 0 0 1 1 0 0 0 0 1 1 0 0 1 1 0 0 0 1 0 0 1 0 0 1 1 1 1 1 1 1
 0 1 1 0 0 0 1 0 1 0 0 0 0 1 0 0 1]
```

In [36]:

```
print(classification_report(y_test,dt_pred))
print("roc_auc score: ",roc_auc_score(y_test,dt_pred))
dt_score = accuracy_score(y_test,dt_pred)
dt_recall = recall_score(y_test,dt_pred)
```

```
              precision    recall  f1-score   support

           0       0.67      0.80      0.73        41
           1       0.81      0.68      0.74        50

    accuracy                           0.74        91
   macro avg       0.74      0.74      0.74        91
weighted avg       0.75      0.74      0.74        91

roc_auc score:  0.7424390243902441
```

In [37]:

```
cm_1 = confusion_matrix(y_test, dt_pred)
plt.figure()
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion matix - Decision Tree Classifier")
sns.heatmap(cm_1, annot = True)
plt.show()
```

After implementing Decision Tree i get accuracy 70%, roc_auc_score 71%, recall 64%. that means Logistic Regrassion is performing well

Hyperparameter tunning on Decision tree

In [38]:
```
from sklearn.model_selection import GridSearchCV
```

In [39]:
```
param_grid = {
    "max_depth": [3,5,7, None],
    "min_samples_split": [2,5,10],
    "min_samples_leaf": [1,2,4]
}
```

In [40]:
```
dt_grid = GridSearchCV(
    estimator=DecisionTreeClassifier(random_state=42),
    param_grid=param_grid,
    cv=5,
    scoring="recall"
)
```

In [41]:
```
dt_grid.fit(x_train,y_train)
```

Out[41]:
```
▸ GridSearchCV
▸ best_estimator_: DecisionTreeClassifier
    ▸ DecisionTreeClassifier
```

In [42]:
```
best_dt = dt_grid.best_estimator_
```

In [43]:
```
print("best parameters: ", dt_grid.best_params_)
```

```
best parameters:  {'max_depth': 3, 'min_samples_leaf': 4, 'min_samples_split': 2}
```

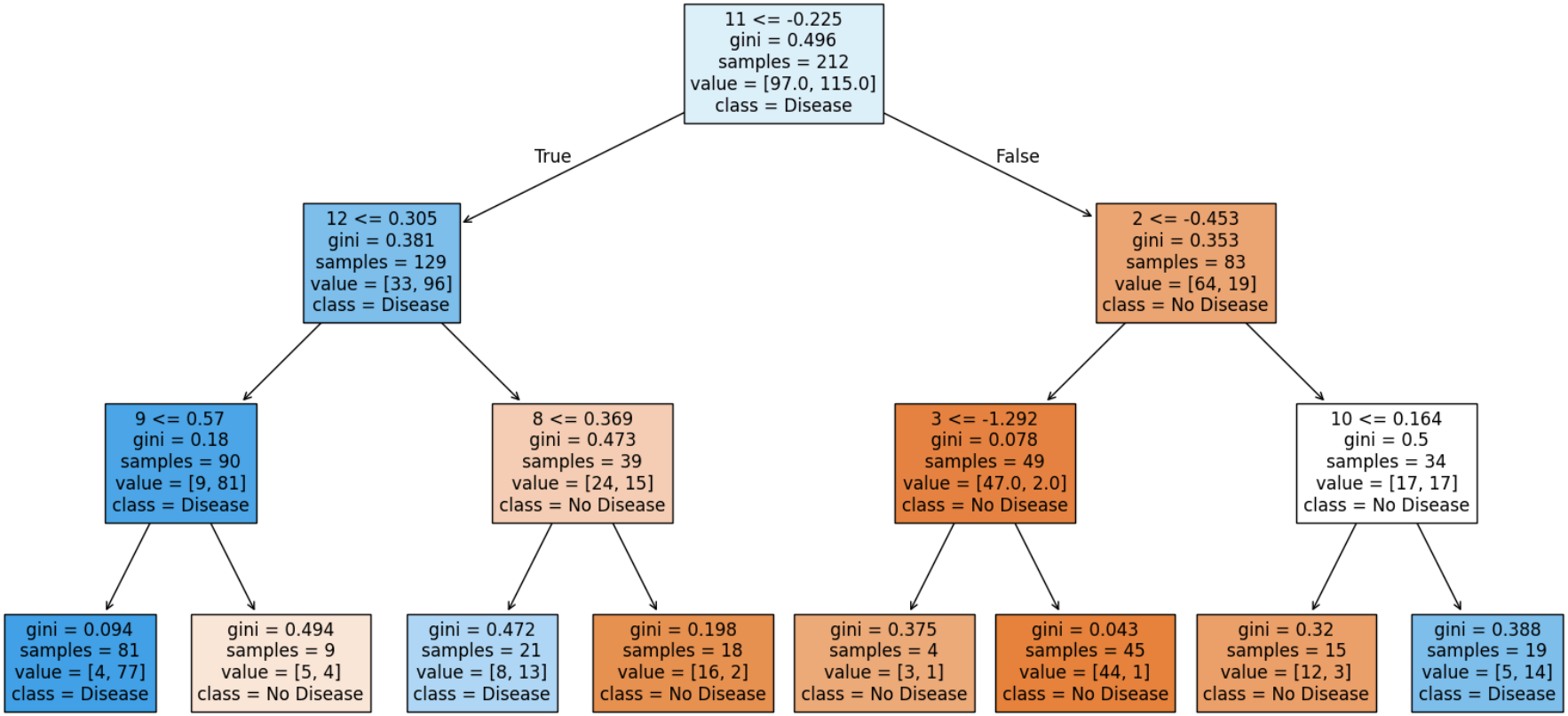
In [44]:
```
y_pred_best_dt = best_dt.predict(x_test)
tunned_recall = recall_score(y_test, y_pred_best_dt)

print("Tuned Decision Tree Accuracy:", tunned_recall)
```

```
Tuned Decision Tree Accuracy: 0.76
```

In [45]:
```
print("Before Tuning recall:", dt_recall)
print("After Tuning recall :", tunned_recall)
```

```
Before Tuning recall: 0.68
After Tuning recall : 0.76
```

In [46]:
```
plt.figure(figsize = (20,10))
plot_tree(best_dt, feature_names = x.columns, class_names = ["No Disease", "Disease"],filled = True)
plt.show()
```



Hyperparameter tuning was applied to the Decision Tree classifier using GridSearchCV with 5-fold cross-validation. Parameters such as maximum depth and minimum samples per split were optimized to reduce overfitting and improve generalization performance.

Random Forest Classifier

In [47]:
```
from sklearn.ensemble import RandomForestClassifier
```

In [48]:
```
rf = RandomForestClassifier()
```

In [49]:
```
# traning
```

```
rf.fit(x_train, y_train)
```

Out[49]:

```
RandomForestClassifier
RandomForestClassifier()
```

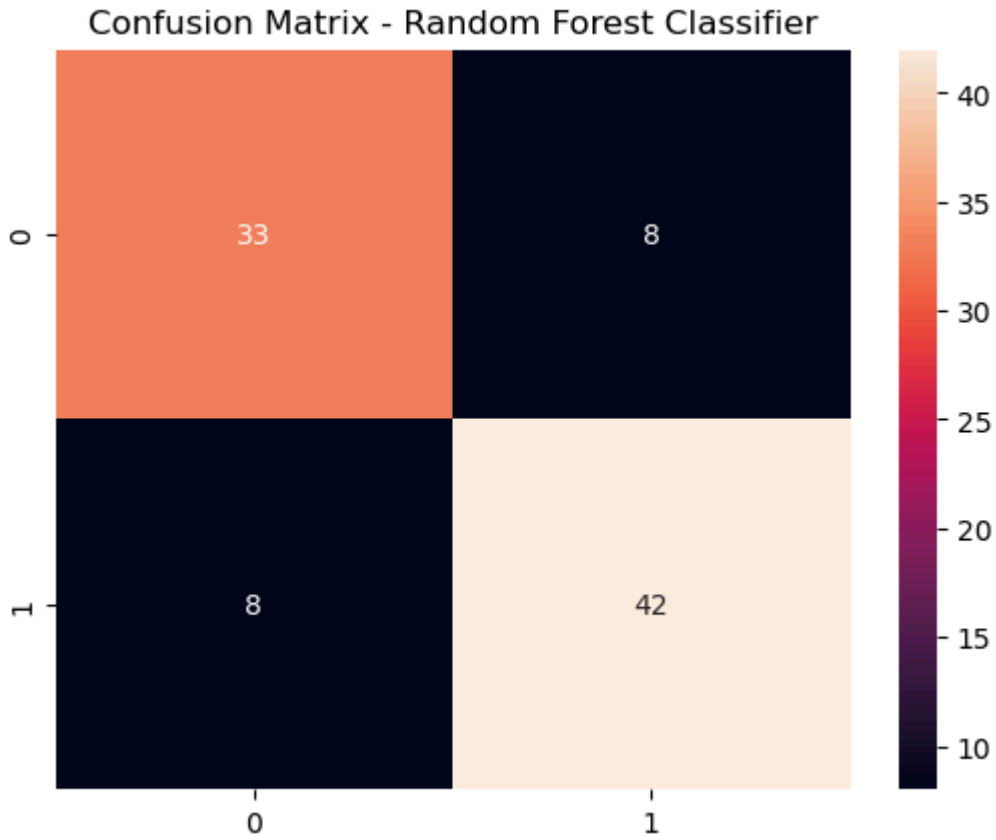
In [50]:

```
rf_pred = rf.predict(x_test)
print(rf_pred)
```

```
[0 1 1 0 1 1 1 0 0 0 1 0 1 0 1 1 1 0 0 0 0 0 1 1 1 1 1 1 0 1 0 0 0 0 1 0 1
 1 1 1 1 1 1 1 1 0 1 1 0 0 0 0 1 1 0 0 0 1 0 0 0 1 0 1 1 0 0 1 1 1 1 1 0 1
 0 1 1 0 0 0 1 0 1 1 0 0 0 1 1 1 1]
```

In [51]:

```
cm_2 = confusion_matrix(y_test, rf_pred)
plt.figure()
plt.xlabel("Predication")
plt.ylabel("Actual")
plt.title("Confusion Matrix - Random Forest Classifier")
sns.heatmap(cm_2, annot = True)
plt.show()
```



In [52]:
```
print(classification_report(y_test,rf_pred))
print("roc_auc score: ",roc_auc_score(y_test,rf_pred))
print()
```

```
              precision    recall  f1-score   support

           0       0.80      0.80      0.80        41
           1       0.84      0.84      0.84        50

    accuracy                           0.82        91
   macro avg       0.82      0.82      0.82        91
weighted avg       0.82      0.82      0.82        91

roc_auc score:  0.8224390243902439
```

Random Forest Classifier is performing better than Logistic Regression and Decision Tree Classifier

HyperParameter tunning on RandomForestClassifier

In [53]:
```
from sklearn.model_selection import RandomizedSearchCV
```

In [54]:
```
param_dist = {
    "n_estimators": [100,200,300],
    "max_depth": [5,10,None],
    "min_samples_split": [2,5,10],
    "min_samples_leaf": [1,2,4]
}
```

In [55]:
```
rf_random = RandomizedSearchCV(
    estimator = RandomForestClassifier(random_state = 42),
    param_distributions = param_dist,
    n_iter=10,
    cv=5,
    scoring="roc_auc",
    random_state=42,
    n_jobs=-1
)
```

In [56]:
```
rf_random.fit(x_train, y_train)
```

Out[56]:

```
▸ RandomizedSearchCV
▸ best_estimator_: RandomForestClassifier
    ▸ RandomForestClassifier
```

In [57]:

```python
best_rf = rf_random.best_estimator_

print("Best Parameters:", rf_random.best_params_)
```

```
Best Parameters: {'n_estimators': 200, 'min_samples_split': 5, 'min_samples_leaf': 4, 'max_depth': 5}
```

In [58]:

```python
from sklearn.metrics import recall_score
y_pred_best_rf = best_rf.predict(x_test)

print("Tuned RF Accuracy:", accuracy_score(y_test, y_pred_best_rf))
print("Tuned RF Recall  :", recall_score(y_test, y_pred_best_rf))
print("Tuned RF ROC-AUC :", roc_auc_score(y_test, y_pred_best_rf))
```

```
Tuned RF Accuracy: 0.8241758241758241
Tuned RF Recall  : 0.86
Tuned RF ROC-AUC : 0.8202439024390243
```

In [59]:

```python
print("----- Comparison -----")
print("Default RF Recall :", recall_score(y_test, rf_pred))
print("Tuned RF Recall   :", recall_score(y_test, y_pred_best_rf))
```

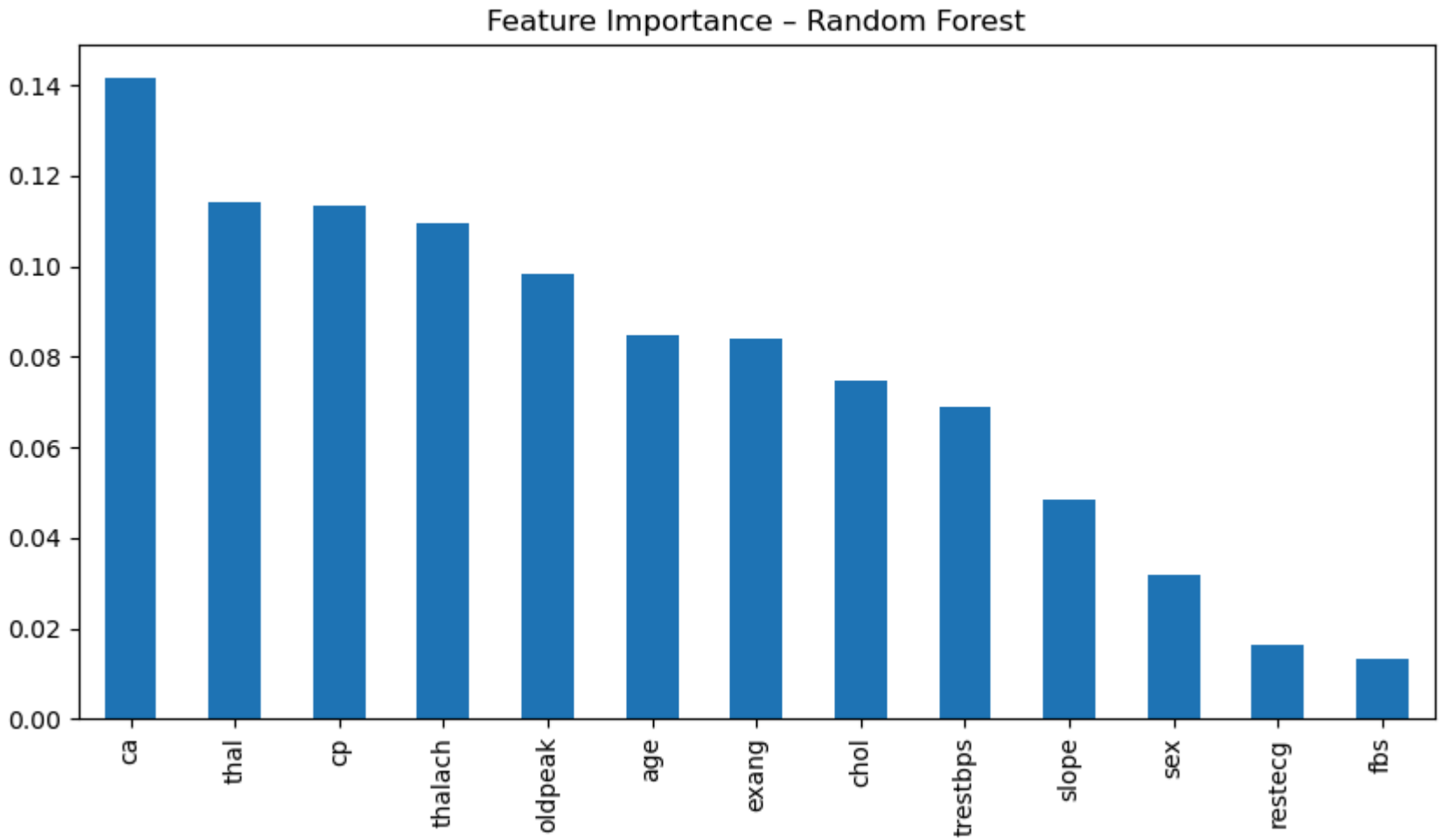
```
----- Comparison -----
Default RF Recall : 0.84
Tuned RF Recall   : 0.86
```

Hyperparameter tuning was applied to the Random Forest classifier using RandomizedSearchCV with 5-fold cross-validation. Parameters such as number of trees, maximum depth, and minimum samples per split were optimized. ROC-AUC was used as the scoring metric to prioritize recall and overall diagnostic performance.

In [60]:

```python
# Helps identify key risk factors for heart disease.
importances = pd.Series(rf.feature_importances_,index=x_duplicate.columns)
importances.sort_values(ascending=False).plot(kind = "bar",figsize=(10,5))
```

```
plt.title("Feature Importance – Random Forest")
plt.show()
```



The Random Forest Classifier model achieved the best performance across all metrics. Due to its higher recall and ROC-AUC score, it is suitable for heart disease prediction where minimizing false negatives is critical.

In [ ]: